STANDARD OPERATING PROCEDURE OF OIL RESISTANCE VF-I05

Lab /SOP/ 55



1.Analytical balance

2.Acetone or isopropyl alcohol

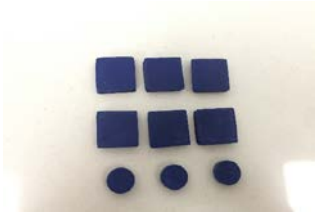
3.Distilled water

4.Blotting paper/filter paper

5.ASTM Fuel B

6.ASTM IRM 903 Oil

7.2,2,4-trimethylpentane (ISO 20344:2011)



Test three (3) samples per material

6.35 mm: solid material for ASTM Fuel B, ASTM IRM 903 Oil

4±0.5 mm: solid material for 2,2,4-trimethylpentane (ISO 20344:2011)

25.4mm x 25.4mm: ASTM Fuel B and ASTM IRM 903 Oil

16 ± 1 mm diameter: 2,2,4-trimethylpentane (ISO 20344)



1.Measure and record the dry weight of each specimen in air before immersion. W1

2.Dip the specimen in acetone and rinse with distilled water.



3.Measure and record the weight of the specimen in distilled water by attaching it to the weighted specimen holder. Measure and record the weight of the specimen holder in water. Calculate the weight of the specimen in water by subtracting the weight of the specimen holder in water from the weight of the specimen and specimen holder in water. W2



4.Immerse the specimens in the desired liquid:

When testing with ASTM Fuel B and ASTM IRM 903 oil, immerse the specimens for 46 ± 0.5 hours at 23°C

When testing with 2,2,4-trimethylpentane (ISO 20344), immerse the specimens for 22 ±0.25 hours at 23°C.

5.Immediately after removal from the immersion medium, quickly dip the specimen in acetone or isopropyl alcohol and blot lightly with filter paper.



6.Measure the weight of specimen in air after immersion: W3

7.Measure the weight of the specimen in distilled water after immersion, Follow procedure in step 3. W4

8.Repeat steps 1- 7 for the other specimens.



Caculation:

$$\% \text{ volume change} = \frac{(W_3 - W_4) - (W_1 - W_2)}{(W_1 - W_2)} \times 100$$

W1 = weight of specimen in air before immersion (g)

W2 = weight of specimen in distilled water before immersion (g)

W3 = weight of specimen in air after immersion (g)

W4 = weight of specimen in distilled water after immersion (g).



Report:

1.% volume change

2.Include material description, temperature, time of immersion, thickness of sample, immersion medium, and any deviations from this procedure

Prepared by	Approved by	REVISION		DATE	